

AI-Powered Criminal Network Analysis System

From piles of paperwork to a map of the criminal world — in seconds.

Smart India Hackathon 2026 · Problem 26189

Ministry of Home Affairs — NCRB, Women Safety Division · Theme: Blockchain & Cybersecurity

The Problem: Investigators Are Drowning in Disconnected Data

Evidence exists — the connections don't

Scattered sources

FIRs, call detail records, bank transfers, prison rosters, travel logs and vehicle data live in separate silos across states and agencies.

Manual cross-checking

An investigator must read every file and connect dots mentally. Cross-state links take weeks of correspondence — or are simply never found.

Networks stay hidden

Kingpins deliberately keep their names out of FIRs. Brokers use unregistered burner phones. Paper-based analysis cannot see them.

The cost

Organised crime operates as a network; investigation happens case-by-case. A Delhi arrest and a Mumbai seizure that share a phone number are treated as unrelated events. The result: street-level dealers are caught, the network above them survives.

Problem 26189 asks for an AI system that analyzes large volumes of criminal and intelligence data to uncover these networks — with accountability built in.

Our Solution: A Six-Stage Intelligence Pipeline

Working prototype — every stage runs live

1 · Ingest

FIR text, CDR, bank, prison, travel, vehicle files (CSV/TXT/PDF)

2 · Extract

NLP pulls names, phones and relationships out of raw FIR narratives

3 · Graph

People become nodes; calls, money, co-accusations, shared cells become edges

4 · Analyze

PageRank finds influence · community detection finds gangs · link prediction finds hidden ties · burst detection finds anomalies

5 · Explain

AI writes plain-language briefs: "X appears to be the broker between Group A and B"

6 · Audit

Every officer query sealed in a blockchain hash chain — tamper-proof by construction

Open-source end to end: Python · FastAPI · NetworkX → Neo4j · React/vis-network

What the Prototype Reveals — Live

One click cracks a two-state case

Two gangs surface instantly

Community detection separates a Delhi drug network and a Mumbai smuggling ring from 21 raw FIRs — no manual tagging.

The broker: a burner phone

One unregistered number talks to both gang leaders. Betweenness centrality flags it as the bridge between the groups.

The hidden kingpin

₹7,00,000 flows through a shell company to a man named in ZERO FIRs. Financially central, operationally invisible — the system flags him in one second.

The origin story

Prison records show the two gang leaders shared Tihar Block-4 in 2022 — the system surfaces how the alliance formed.

Proactive alerting

Anomaly detection catches an 8x spike in calls between the groups — 48 hours before the seizure in the case data. In deployment, spikes near watched nodes trigger real-time alerts to investigators.

Blockchain Where It Actually Matters: Accountability

Theme fit: Blockchain & Cybersecurity — structural, not bolted on

The risk of powerful tools

A system that can map anyone's connections could be misused to surveil innocent citizens. Trust requires that every use of the system is permanently on the record.

Our answer: an immutable audit chain

Every officer query becomes a SHA-256 block sealed by the previous block's hash. Editing ANY past record breaks every later hash. We demonstrate this live with a "Tamper Test" button — the chain instantly reports itself broken.

Production design

Permissioned Hyperledger Fabric replicated across NCRB data centres — no single administrator can rewrite history. Combined with role-based access control and purpose-limited queries, this addresses DPDP Act 2023 obligations head-on. The AI suggests leads with evidence attached; human officers decide. Nothing is a black-box accusation.

Feasibility: Prototype Today, NATGRID-Grade Tomorrow

Every mock maps to a real government source with the same schema

Prototype (runs now)

Synthetic FIR text files

CDR / bank CSVs

Regex entity extraction

NetworkX in-memory graph

Template AI briefs

JSON hash chain

Production (swap connectors, not code)

CCTNS — secure API to national FIR/chargesheet database

Telecom CDR feeds · FIU-IND suspicious transaction reports

IndicBERT NER — FIRs in 12+ Indian languages

Neo4j cluster — billions of edges, same schema

On-prem LLM (air-gapped) fed identical graph facts

Hyperledger Fabric across NCRB nodes

Cost & rollout

100% open-source stack — zero licence cost vs foreign tools (i2, Palantir). Pilot: one district's anonymized historical data, measuring time-to-lead vs manual investigation. Hosted on NIC MeghRaj cloud.

Why This Wins: Uniqueness & Impact

Hidden-kingpin detection

Novel heuristic: high net money inflow + zero FIR mentions + shared covert infrastructure. Finds exactly the people paper-based investigation structurally cannot.

Evidence-cited AI, zero hallucination

Every insight cites its source edge — FIR number, call count, rupee amount. The LLM phrases deterministic graph facts; it never invents connections.

Live tamper-proof audit demo

Judges see the blockchain catch a rogue edit in real time — accountability you can watch working, not a buzzword on a slide.

Weeks → seconds

Cross-state link discovery that takes weeks of file correspondence happens in one click, with the reasoning shown in plain language any officer can act on.

Impact

Investigators stop at street dealers because the network above them is invisible. Making it visible changes what is prosecutable: brokers, financiers and kingpins instead of foot soldiers. For NCRB and the Women Safety Division, the same engine maps trafficking networks — where speed of connection-finding directly protects victims.